

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-1997

Reynolds Hill 138 kV
April 01, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-1997. The Interconnection Customer proposes a 34.6 MW battery energy storage facility interconnecting at the Reynolds Hill 138 kV substation in the Kentucky area. The request was received on February 05, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 438 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,480,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
SUNSET N to PRAGER 138.0 kV	OSAWATOMIE to WAGNER LAKE SUBSTATION 230.0 kV	105.9	237	8.4
BRUNSWICK CELLULOSE SUBSTATION to SHANNON FARM 138.0 kV line	COOPER to UNIVERSAL STUDIOS TAP 138.0 kV	104.9	322	9.5
BRUNSWICK CELULOSE to BOZEMAN SOUTH 138.0 kV	UNIVERSAL STUDIOS TAP to COOPER 138.0 kV line	104.8	304	9.5
BEAVER VLY SUBSTATION to AES BEAVER VLY PARTNERS BEAVER VLY 500.0 kV	CRANBERRY to BEAVER VALEY SUBSTATION 500.0 kV	100.5	2,786	7.1
HOWLAND RD to REYNOLDS HILL 138.0 kV	REYNOLDS HILL to FT RUCKER 138.0 KV L	100.0	222	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SUNSET N to PRAGER 138.0 kV	Reconductor line	\$3,324,000	1% (\$44,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BRUNSWICK CELLULOSE SUBSTATION to SHANNON FARM 138.0 kV line	Reconductor line	\$2,492,000	5% (\$121,000)
BRUNSWICK CELLULOSE to BOZEMAN SOUTH 138.0 kV	Reconductor line	\$3,112,000	5% (\$151,000)
BEAVER VLY SUBSTATION to AES BEAVER VLY PARTNERS BEAVER VLY 500.0 kV	Reconductor line	\$9,317,000	2% (\$145,000)
HOWLAND RD to REYNOLDS HILL 138.0 kV	Reconductor line	\$10,019,000	100% (\$10,019,000)
	Total allocated to project		\$10,480,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.